

Homework Assignment

Fall 2026 Embedded Systems (CEG-7360-01)

Due Date: September 18, 2026

Part 1 — LUT-Based Ring Oscillator

- From **Slide 30**, implement the **LUT-based ring oscillator** using the transistor/switching concept discussed in class.
- Program the required **LUTs** and generate the ring-oscillator signal.
- Simulate the behavior and verify the **oscillation and clock waveform**.
- Determine the required **LUT configuration and control signals**.

Part 2 — Six-Transistor Switching Network

Consider the six-transistor switching network discussed in class:

- Two of the outer transistors are **PMOS** and two are **NMOS**.
- The two inner transistors consist of **one PMOS and one NMOS**.
- Signals can be routed between **N, S, E, and W**.

Program the transistor switching network and determine the required **ON/OFF transistor states and control signals** to route data between CLBs:

- **E** → **W**
- **N** → **S**
- **E** → **N**, via the **shortest path**
- **W** → **S**, via the **shortest path**

For each case:

- a. Identify the **routing path**.
- b. Determine the **ON/OFF state of all six transistors**.
- c. Write the required **control signals**.
- d. Implement the switching logic in **VHDL or Verilog**.
- e. Verify each routing operation using **Vivado simulation** and include the **waveforms**.

Part 3 — Homework

Priority: Complete all questions.

1. **Implement and Simulate:**

Implement and simulate all examples from **Lectures 49–59**.

2. **HDL Code:**

Write the corresponding **VHDL or Verilog** code for each example.

3. **Vivado Simulation:**

Simulate all designs using **AMD Xilinx Vivado** and verify the results.

4. **Report:**

Prepare a short report describing the **inputs, outputs, code, simulation results, waveforms, and observations**.

Submission

Submit the completed homework through the course page on Pilot.